TU Graz

SCIENCE
PASSION
TECHNOLOGY

Data Integration and Large-Scale Analysis: Introduction & Course Overview

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Version 1.0.0

Outline

1 Motivation

2 Administrative Information

3 Course Organization

Motivation

Why is this course relevant?

Illustrative Example

- Consider the case of patient data
  - Need to combine multiple sources
    - Electronic health records
    - Reports, e.g., discharge reports
    - Images, e.g., radiology
    - Population data, e.g., genome sequences
    - Wearable devices

Illustrative Example - Challenges

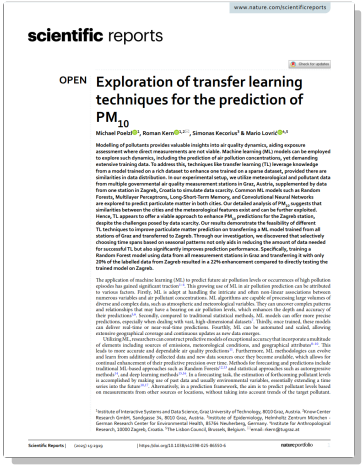
- Data Integration
  - Heterogeneity
    - Data formats, structures
  - Data Quality
    - Outliers, missing values
  - Semantics
    - ICD-10 vs. SNOMED CT
    - “mg/dl” vs. “mmol/l”
  - Disambiguation
    - KAGES stands for Kentucky Association for Gifted Education?
- Large-scale Analysis
  - Volume
    - Number of records/images/...
  - Velocity
    - Constant stream of updates
  - Data Protection
    - Anonymisation (GDPR)
    - Data remains on-premises
  - Analytics
    - Machine Learning / Data Science on distributed datasets

Particulate Matter

- Data Integration
  - Normalise data sources
  - Combine data sources
  - Single schema
  - Unified access
- Large-scale Analysis
  - Efficient data storage & retrieval
  - Analysis of large datasets
  - Data stream processing

Course Content

- Publication
  - Transfer learning of PM10 (particulate matter)
  - Graz → Zagreb
  - Currently working on follow-up publication
    - Taking temperature, altitute, etc into account



www.nature.com/scientificreports

scientific reports

Check for updates

OPEN

Exploration of transfer learning techniques for the prediction of PM_{10}

Michael Poeltl[1], Roman Kern[2,3*], Simonae Kecorius[4] & Mario Lovrić[5,6,7]

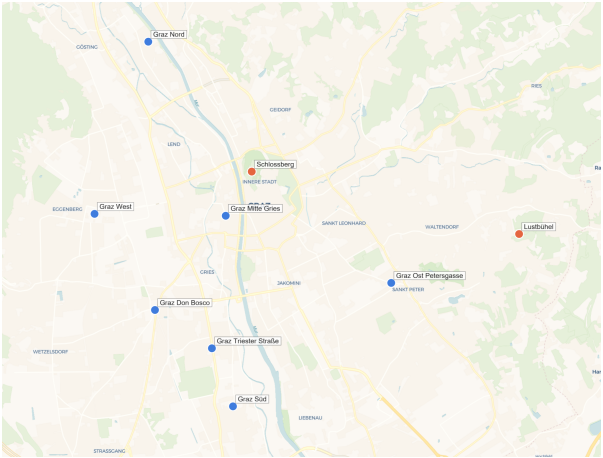
Modelling of pollutants provides valuable insights into air quality dynamics, aiding exposure assessment where direct measurements are not viable. Machine learning (ML) models can be employed to explore such dynamics, including the prediction of air-pollution concentrations, yet demanding extensive training data. To address this, techniques like transfer learning (TL) leverage knowledge from a model trained on a rich dataset to enhance one trained on a sparse dataset, provided there are similarities in data distribution. In our experimental setup, we utilize meteorological and pollutant data from multiple governmental air-quality measurement stations in Graz, Austria, supplemented by data from one station in Zagreb, Croatia to simulate data scarcity. Common ML models such as Random Forests, Multilayer Perceptrons, Long-Short-Term Memory, and Convolutional Neural Networks are explored to predict particulate matter in both cities. Our detailed analysis of PM_{10} suggests that selectively choosing time spans based on seasonal patterns not only aids in reducing the amount of data needed for successful TL but also significantly improves prediction performance. Specifically, training a Random Forest model using data from all measurement stations in Graz and transferring it with only 20% of the labelled data from Zagreb resulted in a 22% enhancement compared to directly testing the trained model on Zagreb.

The application of machine learning (ML) to predict future air pollution levels or occurrences of high-pollution episodes has gained significant traction[1–3]. This growing use of ML in air pollution prediction can be attributed to various factors. Firstly, ML is adept at handling the intricate and often non-linear associations between numerous variables and air pollutant concentrations. ML algorithms are capable of processing large volumes of diverse and complex data, such as atmospheric and meteorological variables. They can uncover complex patterns and relationships that may have a bearing on air pollution levels, which enhances the depth and accuracy of their predictions[4,5]. Secondly, compared to traditional statistical methods, ML models can offer more precise predictions, especially when dealing with large datasets[6]. Moreover, ML models can be trained to operate in near-real-time or near-real-time predictions. Fourthly, ML can be automated and scaled, allowing extensive geographical coverage and continuous updates as new data emerges.

Utilizing ML, researchers can construct predictive models of exceptional accuracy that incorporate a multitude of elements including sources of emissions, meteorological conditions, and geographical attributes[4,6]. This leads to more accurate and dependable air quality predictions[7]. Furthermore, ML methodologies can evolve and learn from additionally collected data and new data sources over they become available, which allows for continuous improvement of the model's performance[8]. This is a significant advantage over static models and traditional ML-based approaches such as Random Forests[9,10] and statistical approaches such as autoregressive methods[11], and deep learning methods[12,13]. In a forecasting task, the estimation of forthcoming pollutant levels is accomplished by making use of past data and usually environmental variables, essentially extending a time-series into the future[14,15]. Alternatively, in a prediction framework, the aim is to predict pollutant levels based on measurements from other sources or locations, without taking into account trends of the target pollutant.

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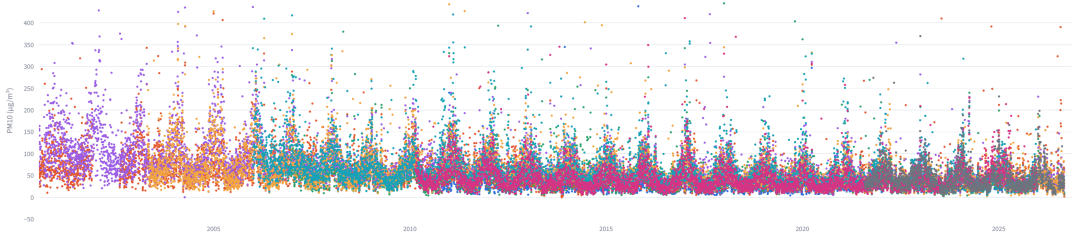
Particulate Matter - Measurement Stations



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Motivation

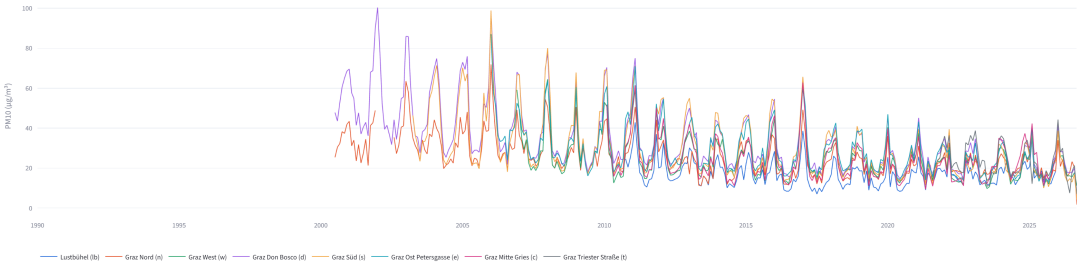
Particulate Matter - Raw Data



Feifei Niu; Roman Kern; Denis Kalkofen, SAI; IML; IVC

Motivation

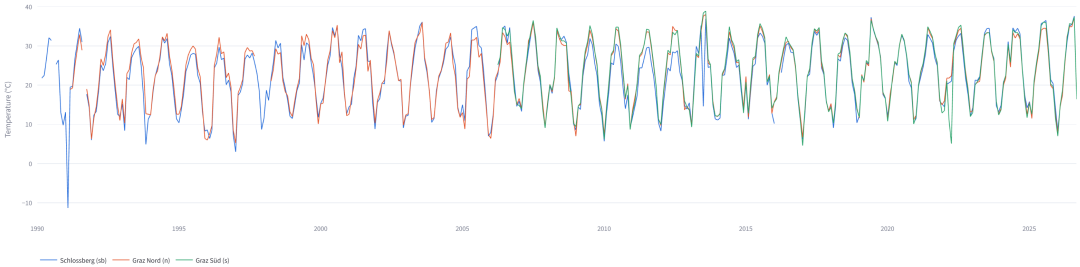
Particulate Matter - Monthly Data



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Motivation

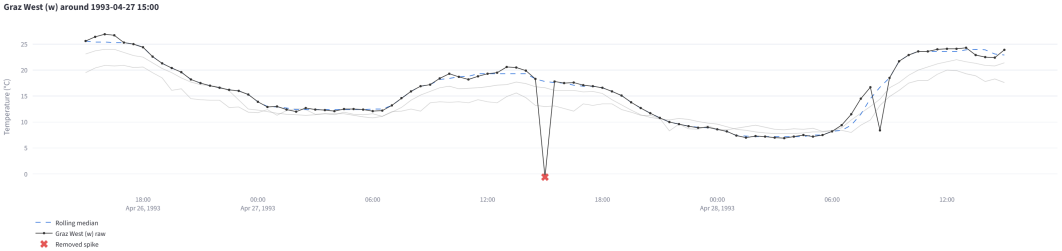
Particulate Matter - Temperatures



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Motivation

Particulate Matter - Outliers



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Administrative Information

About the course

TUG Online

Enroll to the course

- DAT.B32UF 26W 3SSt VU Data Integration and Large-Scale Analysis
- Deadline: 25.03.26, 23:59

TUG Online

Attend the course

- Lecture with integrated exercises (VU)
- Total of 5 ECTS
- (Bi-)weekly lectures
  - Friday, 15:00 - 17:00
  - HS i5
- Attendance is not required

Prequisites

- Databases, Data Management
  - Basic understanding of SQL and relational algebra
- Basic programming skills
  - e.g., Python, R

Course Organisation

When & What

Course Calendar

- Please register until the 25.10.2026
- The course will take place
  - Friday, 15:00 - 17:00 (s.t.)
  - In HS i5 with stream/recording in TUBe

Course Calendar - Part 1

- 02.10. - Course Organization
- 16.10. - Data Type & Structures
- 23.10. - Schema & Synthetic Data
- 06.11. - Ontology Alignment, Entity Linking
- 20.11. - Data Lake, Data Spaces, Data Warehous

Course Calendar - Part 2

- 27.11. - Large Scale Visual Analysis I
- 04.12. - Large Scale Visual Analysis II
- 11.12. - Large Scale Visual Analysis III

Course Calendar - Part 3

- 18.12. - Cloud Resource Scheduling
- 15.01. - Distributed Data Storage and Computation
- 22.01. - Stream Processing
- 29.01. - Exam

Important Deadlines

- 20.01. - Assignment 1: Data Integration
- 31.01. - Assignment 2: Visual Data Analysis

Course Logistics

- TeachCenter
  - Lecture videos
  - Slides and further resources
  - Deadlines and assignments

Thank You!

... for your attention